

Scholastic Aptitude Test, 2017 EC, natural science, entrance exam, round 1

Grade 12 · Aptitude · 2017 E.C.

Time allowed: 120 minutes

Questions: 50

Total marks: 50

Name _____ Grade _____ ID No. _____ Date _____

1. KEY: LOCK; _____ : Computer (1)

A. Monitor	B. Password
C. Server	D. Screen
2. PHYSICIAN: HOSPITAL; _____ : School (1)

A. Supervisor	B. Teacher
C. Administrator	D. Student
3. NEST: BIRD; _____ : Lion (1)

A. Hive	B. Cave
C. Backyard	D. Den
4. AUTOMOBILE: GARAGE; _____ : Hangar (1)

A. Train	B. Airplane
C. Deck	D. Truck
5. PHONEY (1)

A. Genuine	B. Authentic
C. Fake	D. Funny
6. THEFT (1)

A. Gossip	B. Suspicion
C. Burglary	D. Hypocrisy
7. HUMOROUS (1)

A. Amusing	B. Doubting
C. Satiating	D. Satisfying
8. INVINCIBLE (1)

A. Visible	B. Conquerable
C. Breakable	D. Debatable
9. FONDLY (1)

A. Hatred	B. Agreeable
C. Affectionate	D. Sociable
10. SUBORDINATE (1)

A. Soldier	B. Inferior
C. Superior	D. Retainer

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11. The government has just repealed the law segregating public facilities so that there will be no longer discrimination (1)
- A. signed
C. approved
- B. cancelled
D. renewed
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12. The man was killed in a car accident. His body was dragged from the burning wreckage of his car. (1)
- A. backseats
C. remains
- B. wheels
D. fire
-
13. Harvard is one of the most prestigious universities in the USA. (1)
- A. furnished
C. privileged
- B. expensive
D. admired
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14. Identify the correct sentence: (1)
- A. One of all-time greatest payer has recently retired from football.
B. One of all-time greatest payers have recently retired from football.
C. One of all-time greatest payer have recently retired from football.
D. One of all-time greatest payers has recently retired from football.
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15. Identify the correct sentence: (1)
- A. I wanted to first take shower and then ate snack.
B. I wanted to first take shower and then eating snack.
C. I wanted to first take shower and then eat snack.
D. I wanted first take shower and then eat snack.
-
16. Identify the correct sentence: (1)
- A. Their school takes only twenty minutes on foot.
B. They're school takes only twenty minutes on foot.
C. There school takes only twenty minutes on foot.
D. Theirs school takes only twenty minutes on foot.
-
17. Identify the correct sentence: (1)
- A. If you'll not feel well, you better go to clinic.
B. If you're not feeling well, you'd better go to clinic.
C. If you'll not feel well, you'd better go to clinic.
D. If you're not feeling well, you better go to clinic.
-
18. The main intent of the writer in the text is to imply that _____. (1)
- A. most successful employers are those who have achieved their dreams through trade-off.
B. there are steps that one needs to undergo in the transition from employee to employer.
C. the transition from employee to employer does not require much effort except having the interest.
D. smooth transition from being an employer to employee is more of a matter of destiny than hard work.

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19. What does the sentence most successful employers today have risen through the ranks, relying on their ability to manage and direct others effectively imply? (1)
- A. All successful employers today have reached their level passing through lots of ups and downs.
 - B. Most successful employers have proven their leadership ability through their journey.
 - C. Every successful employer is the result of his/her hard work and diligence.
 - D. Having different ranks and rewards is the real manifestation of successful employers.
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20. In Paragraph 2, the phrase these two individuals refers to _____. (1)
- A. the employer and the employee.
 - B. people and the community as a whole.
 - C. John Smith & the person in movie industry.
 - D. the grocery store and the movie theatre.
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21. According to the passage, what does the word vices most likely denote? (1)
- A. weaknesses
 - B. ambitions
 - C. deputies
 - D. strengths
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22. Why does the author emphasize the act of reflection? (1)
- A. To improve intellectual superiority.
 - B. To make reading more enjoyable.
 - C. To complete the reading process.
 - D. To simplify the ideas encountered.
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23. Which of the ideas is used as evidence for your answer to question number 28? (1)
- A. Line 4 (tired after reading... not in us.)
 - B. Line 1 (What is the ... preoccupation.)
 - C. Line 10 (The assumption ... absurdity.)
 - D. Line 12 (Reading without... ridiculous.)
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24. Which of the following can be rearranged into a 5-letter English word? (1)
- A. GOFAT
 - B. PASEH
 - C. PYRIO
 - D. DWAEP
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25. If two days after tomorrow is four days before Saturday, what day is it today? (1)
- A. Sunday
 - B. Monday
 - C. Friday
 - D. Saturday
-
26. Aki is Bali's sister. Yaya is Bali's mother. Lema is Yaya's father. Gette is Lema's mother. Then how is Aki related to Lema? (1)
- A. grandson
 - B. grandmother
 - C. daughter
 - D. granddaughter
-
27. Statement I: All football players are sportspersons. Statement II: All sportspersons are fit. Conclusion: Some football players are not fit. This conclusion is: (1)
- A. reasonably correct
 - B. appropriate
 - C. incorrect
 - D. can't be determined
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28. How many months have a twenty-eighth day? (1)
- A. one month
 - B. two months
 - C. seven months
 - D. twelve months

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29. Gad is older than Dan and Dan is older than Betti. With Senni being older than Dan and younger than Gad, who is the youngest? (1)
- A. Gad
B. Dan
C. Betti
D. Senni
-
30. Which one of the following statements is true? (1)
- A. Every non-negative integer is a natural number.
B. Every integer number can be numerator of a rational number.
C. The sum of two irrational numbers is irrational number.
D. The intersection of the set of integers and the set of irrational numbers is non-empty set.
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31. Which one of the following is the largest number? (1)
- A. $\sqrt[3]{2}$
B. $\sqrt[6]{5}$
C. $\sqrt[4]{3}$
D. $\sqrt[12]{11}$
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32. A ball is shot into the air from the top of the building. The height in meter is given by $h(t) = kt - t^2 + 30$, where t is the time taken in second. What is the possible value of k if the maximum value of height from the ground is 34? (1)
- A. 4
B. $\sqrt{2}$
C. 2
D. 5
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33. If $f(x) = \sin \pi x + e^{x^2} + |x| + x^3$ and $g(x) = \cos \pi x + \ln(x + 1) + x$, then what is the value of $f(1) - 2g(0)$? (1)
- A. $\ln 2$
B. e
C. $e + 2$
D. 0
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34. Which one of the following is the domain of $g(x) = \ln(1 - x) + \ln x$? (1)
- A. $\mathbb{R} \setminus \{-1, 0\}$
B. $(0, \infty)$
C. $(1, \infty)$
D. $(0, 1)$
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35. If $-x + 2y = 4$, then what is the value of $\frac{25^y}{5^x}$? (1)
- A. 625
B. 125
C. 25
D. 5
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36. A farmer has ten cows. Every day, he receives the same amount of milk in liters from each cow. One day, due to the food supply, he received half of the daily amount of milk from the four cows; he received three-fourth of the daily amount of milk from the three cows, and no milk from the other three cows. In total, he didn't receive ninety-two liters of milk on that day. How many liters of milk did he receive on that particular day? (1)
- A. 16
B. 92
C. 68
D. 160
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37. A woman burns 416 calories in an hour exercise at a fitness center. If she burns 10 calories per minute swimming in the water and 6 calories per minute pedaling on the stationary bike, then how many minutes of the hour does she spend exercising in the water? (1)
- A. 14
B. 56
C. 46
D. 60

38. A daily profit earned by a T-shirt factory is modeled by the formula $p(x) = -x^2 + 1,400x - 80,000$, where p is the profit and x is the price for each T-shirt sold. What range of prices will yield a profit of at least Birr 400,000? (1)
- A. $x \geq 800$ B. $600 \leq x \leq 800$
C. $x \leq 600$ D. $x \leq 600$ or $x \geq 800$
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39. Which one of the following is true about the function $f(x) = x^{\frac{2}{5}}$? (1)
- A. The range of f is $\{y \in \mathbb{R} \mid y \geq 0\}$ B. The domain of f is $\{x \in \mathbb{R} \mid x \geq 0\}$
C. The domain of f is $\{x \in \mathbb{R} \mid x \leq 0\}$ D. The range of f is $\{y \mid y \in \mathbb{R}\}$
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40. What is the value of z in the sequence 1, 2, 4, 8, 10, 20, 22, z , ...? (1)
- A. 24 B. 44
C. 40 D. 32
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41. What is the possible value of the missing term x in the sequence $-1, 0, 1, 2, -1.8, 1, 26, -1, x, 1, 242, -1, 728, 1, \dots$? (1)
- A. 36 B. 54
C. 34 D. 80
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42. How many days are there from the first day of March to the last day of May? (1)
- A. 92 B. 89
C. 90 D. 91
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43. Consider the lengths of the sides of a triangle x, y, z , and the quantities given below: Quantity P Quantity Q (1)
- $3x + y - z$ $4x$
- A. Quantity P is greater than Quantity Q. B. Quantity P is less than Quantity Q.
C. The two quantities are equal. D. The two quantities cannot be compared.
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44. A corporation pays ten employees a total of Birr 100,000 every month. Six out of ten employees receives less than Birr 10,000 per month. Quantity P Quantity Q The average of the monthly payment of the corporation The median of the monthly payment of the corporation (1)
- A. Quantity P is less than Quantity Q. B. Quantity P is greater than Quantity Q.
C. The two quantities are equal. D. The two quantities cannot be compared.
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45. An engineer is putting black and white colored ceramics on a square floor of area 36m^2 . Each ceramic has area 1m^2 . The engineer places the black ones along both diagonals and the rest is covered by white colored ceramics. In what ratio is black to white ceramic used? (1)
- A. 1:2 B. 1:6
C. 4:5 D. 1:5
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46. A person makes a glass of juice and used half of the glass with avocado, one third of the remaining half with mangoes and the rest is filled by banana of bananas to avocados in the glass? (1)
- A. 1:2 B. 1:6
C. 2:3 D. 1:3

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47. Two pipes A and B are used to fill a water tank. Pipe A can fill the tank in 3 hours, and pipe B can fill the same tank in 6 hours. How long will it take in hours to fill the tank if both pipes are opened at the same time? (1)
- A. 4
B. 5
C. 2
D. 9
-
48. A milk factory has only two pipe machines used to fill bottles for market. The old and new machines can fill 35 and 50 bottles per hour, respectively, and a total of 590 bottles are filled every day. One day the old machine was broken after working for 4 hours. How long will it take in hours for the new machine to fill the remaining bottles on that day? (1)
- A. 5
B. 8
C. 7
D. 13
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49. If a spherical soccer ball has a volume of $36\pi \text{ cm}^3$, then which one of the following is the surface area of the ball in square centimeter? (1)
- A. 12π
B. $\frac{16\pi}{3}$
C. $\frac{32\pi}{3}$
D. 36π
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50. Points A, B and C are located on a circle with center O and radius 2 units long. If A, B and C are joined to form an equilateral triangle, then what is the area of triangle ABC in square units? (1)
- A. $\sqrt{3}$
B. 3
C. $3\sqrt{3}$
D. 1

— END OF EXAMINATION —